

5.13.16

AI25BTECH11030 -Sarvesh Tamgade

Question: Let k be an integer such that the triangle with vertices

$$\mathbf{A} = \begin{pmatrix} k \\ -3k \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 5 \\ k \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} -k \\ 2 \end{pmatrix}$$

has area 28 sq. units. Find the orthocenter of the triangle.

Solution:

$$\text{Area} = \frac{1}{2} |\mathbf{M}| = 28 \quad (0.1)$$

$$\text{where } \mathbf{M} = (\mathbf{B} - \mathbf{A} \quad \mathbf{C} - \mathbf{A}) = \begin{pmatrix} 5-k & -2k \\ 4k & 2+3k \end{pmatrix} \quad (0.2)$$

$$\det(\mathbf{M}) = (5-k)(2+3k) - (-2k)(4k) = 10 + 13k + 5k^2 \quad (0.3)$$

$$10 + 13k + 5k^2 = \pm 56 \quad (0.4)$$

After solving,

$$k = 2 \quad (0.5)$$

$$\text{For } k = 2, \quad \mathbf{A} = \begin{pmatrix} 2 \\ -6 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} -2 \\ 2 \end{pmatrix} \quad (0.6)$$

The equation of the altitude through $\mathbf{A}$ is:

$$\mathbf{n}^T \mathbf{x} = 1 \quad (0.7)$$

This altitude passes through $\mathbf{A}$, so

$$\mathbf{n}^T \mathbf{A} = 1 \quad (0.8)$$

$$\mathbf{n}^T \begin{pmatrix} 2 \\ -6 \end{pmatrix} = 1 \quad (0.9)$$

and the normal vector $\mathbf{n}$ of the altitude must be parallel to $\mathbf{B} - \mathbf{C}$:

$$\mathbf{n} = t \begin{pmatrix} 7 \\ 0 \end{pmatrix} \quad (0.10)$$

Substitute to get:

$$t \begin{pmatrix} 7 \\ 0 \end{pmatrix}^T \begin{pmatrix} 2 \\ -6 \end{pmatrix} = 1 \implies 14t = 1 \implies t = \frac{1}{14} \quad (0.11)$$

Hence,

$$\mathbf{n} = \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix} \quad (0.12)$$

The equation of altitude through **A** is:

$$\begin{pmatrix} \frac{1}{2} & 0 \end{pmatrix} \mathbf{x} = 1 \quad (0.13)$$

The equation of the altitude through **B** is:

$$\mathbf{n}^T \mathbf{x} = 1 \quad (0.14)$$

This altitude passes through **B**, so

$$\mathbf{n}^T \mathbf{B} = 1 \quad (0.15)$$

$$\mathbf{n}^T \begin{pmatrix} 5 \\ 2 \end{pmatrix} = 1 \quad (0.16)$$

and the normal vector **n** must be parallel to **A** – **C**:

$$\mathbf{n} = t \begin{pmatrix} 4 \\ -8 \end{pmatrix} \quad (0.17)$$

Substitute to get:

$$t \begin{pmatrix} 4 \\ -8 \end{pmatrix}^T \begin{pmatrix} 5 \\ 2 \end{pmatrix} = 1 \implies t(20 - 16) = 1 \implies 4t = 1 \implies t = \frac{1}{4} \quad (0.18)$$

Hence,

$$\mathbf{n} = \begin{pmatrix} 1 \\ -2 \end{pmatrix} \quad (0.19)$$

The equation of altitude through **B** is:

$$\begin{pmatrix} 1 & -2 \end{pmatrix} \mathbf{x} = 1 \quad (0.20)$$

Combining equations (0.13) and (0.20):

$$\begin{pmatrix} \frac{1}{2} & 0 \\ 1 & -2 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (0.21)$$

Writing the augmented matrix:

$$\left(\begin{array}{cc|c} \frac{1}{2} & 0 & 1 \\ 1 & -2 & 1 \end{array} \right) \quad (0.22)$$

Perform the row operation $R_2 \rightarrow R_2 - 2R_1$:

$$\left(\begin{array}{cc|c} \frac{1}{2} & 0 & 1 \\ 0 & -2 & -1 \end{array} \right) \quad (0.23)$$

Therefore, the solution vector is:

$$\mathbf{x} = \begin{pmatrix} 2 \\ \frac{1}{2} \end{pmatrix} \quad (0.24)$$

Answer: The orthocentre of the triangle is

$$\mathbf{x} = \begin{pmatrix} 2 \\ \frac{1}{2} \end{pmatrix}$$

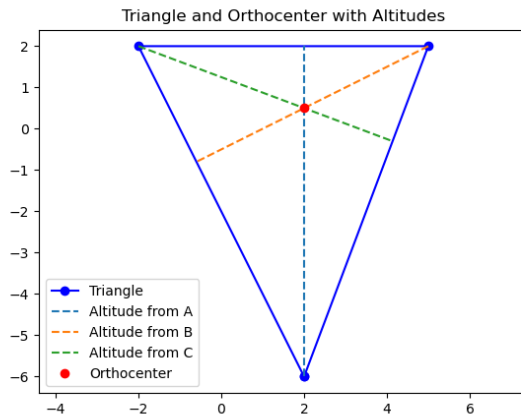


Fig. 0.1: Vector Representation